

# BA2-T3 - Cross-corpus regression of operator/role/modifier candidate and semantic-key/lexical separation

Revision: R1

Status: COMPLETED / PROVISIONAL PASS WITH REDUCTION / BA2 NOT CLOSED

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Phase: BA2 - Relations and canonical action vocabulary

BA0: CLOSED

BA1: CLOSED

BA3: NOT STARTED

## 1. Question

BA2-T3 asks only:

Can the BA2-T2 semantic-operator/role/modifier candidate replay both governed corpora without raw-prose reconstruction, and which candidate distinctions can be removed or promoted while preserving the semantics needed by methodology-neutral consumers?

This is a regression/reduction step. It does not reward vocabulary growth.

## 2. Evidence replayed

The regression uses:

- closed BA1 identity ontology (BAReferent + BAProposition);
- BA2-T1 n-ary proposition structure;
- BA2-T2 R1 vocabulary candidate;
- facial-access D-3.3, FR-3.3, FR-3.4, D-3.4, D-3.5 and SR-3.4-C/I/P;
- order-fulfillment MR-1 through MR-4, including reservation, payment, fulfillment, provider contracts, lifecycle and physical handoff reuse;
- BA0-T2 STRIDE-oriented transfer projection only as a bounded consumer probe.

No source document is changed by this trial.

## 3. Regression strategy

For every candidate operator the test attempts, in order:

1. **delete** it and replay the two corpora;
2. **merge** it into another key and check for semantic loss;
3. **move** the distinction into roles, typed values or constraints where that is more economical;
4. retain the key only if a consumer would otherwise have to reconstruct relevant project meaning from prose.

The same discipline is applied to roles, modifier kinds and classification.

## 4. Operator regression ledger

| BA2-T2 key            | Regression result                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------|
| <b>transfer</b>       | Required for camera delivery and order/provider/carrier conveyance; not equivalent to reference      |
| <b>produce</b>        | Required to preserve governed output/result production; <b>map/derive</b> can use input/result roles |
| <b>create</b>         | Required for OrderEvaluation, Reservation, FulfillmentTask, StockMovement/event establishment        |
| <b>observe</b>        | Required to distinguish reading/querying current state from creation or mutation                     |
| <b>transition</b>     | Required for Order/Reservation lifecycle and state change                                            |
| <b>correlate</b>      | Same-request/evaluation identity matching is stronger than generic reference                         |
| <b>reference</b>      | FulfillmentTask -> Order/Reservation does not imply correlation/dependency                           |
| <b>dependOn</b>       | Required for commitment gates and physical-handoff prerequisites                                     |
| <b>consumeService</b> | Actual service use without ownership transfer is not exhausted by generic dependency                 |

| BA2-T2 key                  | Regression result                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| <b>realize</b>              | Connectivity -> Ethernet/Wi-Fi change must remain mechanically distinct                               |
| <b>assignResponsibility</b> | Required for responsibility/authority placement; can carry an authority kind                          |
| <b>ownOrManage</b>          | Can be represented by <b>assignResponsibility</b> + <b>responsibilityKind</b> + polarity without loss |
| <b>constrain</b>            | Required for security, completion, atomicity, concurrency, idempotency and admissibility rules        |
| <b>classify</b>             | Required for reusable semantic kinds without BA1 subtype proliferation                                |

## Result

BA2-T2 seed keys: 14

BA2-T3 R2 seed keys: 13

missing operator forced by regression: NONE

## 5. Why **ownOrManage** collapses but **consumeService** does not

The facial corpus deliberately states both that the project **consumes** local connectivity and that it does **not own/manage** the underlying transport.

**consumeService** must remain distinct because it says an actual capability/service is used. Ownership/management, however, is one authority mode among possible responsibility modes. It can therefore be represented under the broader responsibility relation:

```

assignResponsibility
  responsibleParty    -> project
  responsibilityScope -> underlying transport
  responsibilityKind  -> ownership/management
  polarity            -> negative

```

The semantic boundary **service consumption != ownership/management** remains visible even though one operator key is removed.

## 6. Missing-role test

Two role distinctions are added because the replay cannot remain explicit without them.

### 6.1 input under produce

**PaymentAdapter** maps **provider response/outcome** to **PaymentAuthorizationResult** and **AvailabilityResult** derives from current accountable inventory state. Merely saying that a result is produced loses the governed input/result boundary.

The regression therefore adds optional **input** participation to **produce**.

### 6.2 **responsibilityKind** under **assignResponsibility**

The merge of ownership/management into responsibility semantics must not make ownership, management and generic responsibility indistinguishable. A controlled **responsibilityKind** term preserves that distinction.

No other missing role distinction is forced by the current corpora.

## 7. Cardinality test

The R2 role matrix uses exact maxima only where the current operator meaning requires them. It avoids inventing general systems-model cardinalities.

Examples:

```

transfer:
  source      1
  destination 1..*
  content     1..*

```

```
correlate:
  correlatedItem      1..*
  correlationContext  1
```

```
transition:
  subject      1
  fromState    0..1
  toState      1
  actor        0..1
```

```
consumeService:
  consumer      1..*
  service       1
  provider      0..1
```

The matrix is internally compatible with both corpora. No clause requires anonymous positional arguments or method-specific DFD slots.

## 8. Lexical-key separation regression

The same semantic key remains stable across distinct source verbs:

- delivery/send/submit/emit pressure -> **transfer** where the governed meaning is conveyance;
- read/query pressure -> **observe**;
- create/record pressure -> **create** only when new project-semantic identity/event is established;
- transition/release/consume/update pressure -> **transition** when governed lifecycle/state changes;
- produce/derive/map pressure -> **produce** with explicit inputs/results and separate constraints when normalization rules matter.

The reverse is also important: identical source words do not force identical BA semantics. **record** may mean create an attributable StockMovement or persist/update an existing state, depending on the governed clause.

### Disposition

```
semantic key stable across source wording: PASS
source lexical predicate as semantic identity: REJECTED
canonical display synonym work: DEFERRED TO BA5
```

## 9. Polarity regression

Negative facts occur in both corpora and do not justify **notX** operator proliferation.

Examples include:

- project does not own/manage transport;
- incomplete transfer is not success;
- stale/superseded results must not satisfy current commitment gates;
- raw provider/carrier status must not directly become governed state;
- rejected/indeterminate carrier acceptance is not physical handoff.

Explicit polarity therefore survives as a normalized proposition property.

**Disposition: RETAIN / READY FOR BA2 CLOSURE REVIEW.**

## 10. Modifier-promotion regression

BA2-T2 allowed a wider modifier candidate set. Replay shows that several of those semantics are too important to remain embedded by default.

### 10.1 Promote to explicit constrain propositions

- completion/failure semantics;
- atomicity;
- concurrency;

- idempotency;
- normative exact-once/all-lines quantity rules when they constrain behavior.

These rules recur, are independently reviewable and are useful selection points for analysis.

## 10.2 Keep only genuinely local modifiers

- `condition` (including local applicability/guard semantics);
- `temporalScope`.

Participant cardinalities are structural role contracts rather than modifiers. If a condition or temporal scope becomes independently reusable/reviewable, the existing promotion rule moves it into referent/proposition semantics.

### Disposition

```
untyped/free-text modifier bag: REJECTED
wide modifier enumeration from T2: REDUCED
condition: RETAIN
temporalScope: RETAIN
atomicity/concurrency/idempotency/completion rules: PROMOTE TO CONSTRAIN
```

## 11. Classification regression

Classification-as-proposition successfully supplies reusable semantic distinctions without turning them into BA1 root types.

A bounded consumer can identify, for example, delivery behavior, RecognitionCapture information/resource meaning and Camera/processor capability meaning from controlled classification assertions rather than from source prose.

The order corpus adds pressure from party, capability, information/resource, store and contract meanings.

However, regression does **not** justify freezing a universal semantic-kind taxonomy. Doing so would recreate the general systems-modeling ontology that BA1 deliberately avoided.

### Disposition

```
classification-as-proposition: RETAIN
stable semantic-kind registry contract: REQUIRED
fixed exhaustive semantic-kind enumeration: REJECTED AS CURRENT-SCOPE REQUIREMENT
semantic kind -> BA1 metaclass promotion: REJECTED
```

## 12. Facial-access replay

The FR-3.4 branch can be selected mechanically as:

Delivery behavior referent

```
transfer(CameraSubsystem, RecognitionProcessor, RecognitionCapture)
correlate(RecognitionCapture, RecognitionRequest)
```

Connectivity meaning

```
consumeService(project context, LocalConnectivity)
assignResponsibility(project, underlying transport, ownership/management) [negative]
realize(LocalConnectivity, WiredEthernet)
```

Delivery constraints

```
incomplete delivery != successful completion
confidentiality
integrity
authorized provenance
```

A future STRIDE overlay can derive source, destination, information, correlation, external responsibility and realization without introducing DataFlow or TrustBoundary into Base Analysis.

### 13. Order-fulfillment replay

The order corpus does not force new operator keys:

- `requestsCapture`, `requestsVoid`, `requestsStockIssue` can be represented through transfer/service-consumption/dependency semantics rather than copied as operators;
- `maps` is handled by `produce` with input/result plus normalization constraints;
- `increases`, `decreasesReserved`, `releases` and `consumes` are lifecycle/state transition and constraint semantics, not separate canonical verbs;
- exactly-once, all-or-nothing, concurrency and idempotency are explicit constraints;
- the physical handoff remains a reusable event/milestone referent and is not collapsed into carrier acceptance.

The corpus therefore passes without vocabulary growth.

### 14. Bounded method-consumer regression

The BA0-T2 STRIDE-oriented transfer projection remains constructible from R2:

```
behavior      <- classify(delivery, behavior/event)
source        <- transfer.source
destination   <- transfer.destination
information   <- transfer.content + classification
correlation   <- correlate
responsibility <- consumeService + assignResponsibility/polarity
realization   <- realize
failure rule  <- constrain
security rules <- constrain
```

STRIDE/DFD categories remain outside Base Analysis.

**Disposition: PASS.**

### 15. BA1 reopen test

No operator, role, modifier or classification mechanism requires a third first-class semantic identity family.

**Disposition: BA1 REMAINS CLOSED.**

### 16. Trial disposition

|                                           |                                             |
|-------------------------------------------|---------------------------------------------|
| BA2-T1 structural candidate               | RETAINED                                    |
| exact semantic operator key               | RETAINED                                    |
| operator-family facet                     | REMOVED FROM NORMATIVE CORE                 |
| operator seed                             | REDUCED 14 -> 13                            |
| ownOrManage                               | MERGED INTO assignResponsibility            |
| missing operator                          | NONE FORCED                                 |
| operator-scoped role contracts            | PASS / R2 MATRIX                            |
| new role: input                           | FORCED BY REGRESSION                        |
| new role: responsibilityKind              | FORCED BY MERGE                             |
| semantic-key/lexical separation           | PASS                                        |
| explicit polarity                         | RETAIN                                      |
| modifier vocabulary                       | REDUCED                                     |
| atomicity/concurrency/idempotency/failure | PROMOTE TO constrain                        |
| classification-as-proposition             | RETAIN                                      |
| fixed exhaustive semantic-kind taxonomy   | NOT REQUIRED                                |
| bounded STRIDE consumer                   | PASS                                        |
| third BA1 identity family                 | NOT FORCED                                  |
| BA2-T3                                    | COMPLETED / PROVISIONAL PASS WITH REDUCTION |
| BA2                                       | STARTED / NOT CLOSED                        |
| BA3                                       | NOT STARTED                                 |

## 17. Smallest unresolved set

No further discovery microstep is justified by a missing operator or corpus counterexample. What remains is a closure review over the reduced R2 candidate:

1. accept/reject the thirteen-key current-scope seed;
2. accept/reject the operator-role matrix and ownership/management merge;
3. accept/reject polarity and the narrowed modifier boundary;
4. accept/reject the semantic-kind registry contract without a universal enumeration;
5. verify that the resulting BA2 contract creates no BA1 or BA3 responsibility leakage.

## 18. Next authorized microstep

Execute only:

**BA2-T4 - relation/action vocabulary closure review over the regressed R2 candidate.**

BA2-T4 must be a closure review, not another vocabulary-expansion exercise. Do not start BA3, formal STRIDE overlays, Common Finding or implementation work.